SMART INDIA HACKATHON 2026

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TITLE PAGE

Problem Statement ID: SIH26061

Problem Statement Title: AI-Driven Smart Energy Management System for Polar Research Stations

Theme: Clean & Green Technology

PS Category: Software

Team ID: <Team ID>

Team Name: <Team Name>

HIMSHAKTI

Fuel-survivability operating policy for Indian Antarctic stations

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HIMSHAKTI — fuel survivability, not another dashboard

Proposed Solution

- Two-level operating policy for a station with ONE fuel delivery a year
- Outer level: Monte Carlo allocator sets a daily litre allowance that holds P(critical load served to the ship) ≥ 99%
- Inner level: rolling MILP re-optimises every step inside that allowance

Detailed explanation

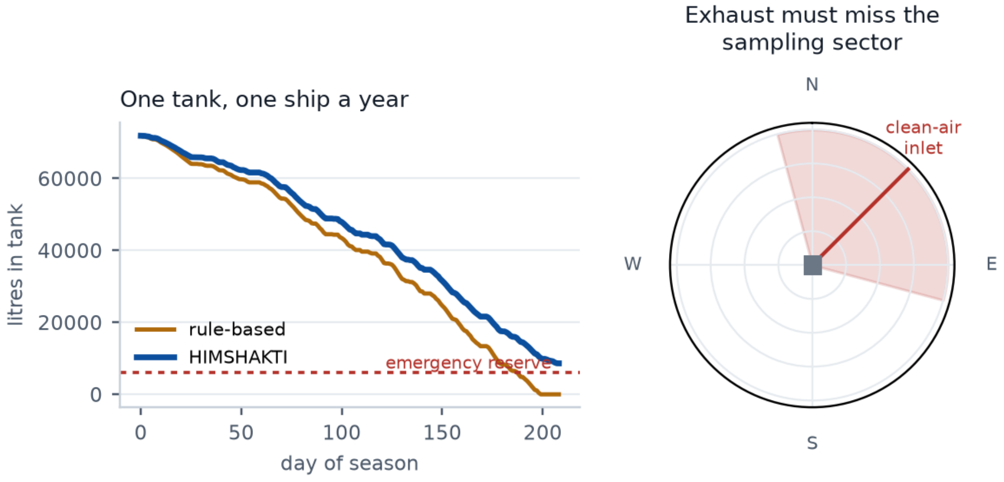
- Physics digital twin of Maitri/Bharati on 9 years of real ERA5 weather
- Loads built bottom-up: headcount, envelope heat loss, snow-melt water, 24x7 science
- PV modelled at 70 deg tilt with snow albedo and cold-temperature gain; wind with air-density gain
- LightGBM quantile forecasts (p10/p50/p90) for load, solar and wind, 1–48 h
- MILP commits gensets, battery, curtailment and tiered shedding on a 36 h horizon

How it addresses the problem

- Delivers all four asks of the statement: load forecasting, renewable integration, fuel optimisation, extreme conditions
- But scores them on survivability to the next ship, not on cost per kWh

Innovation and uniqueness

- Annual-resupply fuel budgeting — absent from every commercial EMS (ABB, Schneider, SMA)
- Science-integrity dispatch — generator hours priced when forecast wind carries exhaust into the station's own clean-air samplers
- Physical-availability forecasting — drifted panels and iced blades, which decorrelate from the weather forecast



Left: same weather, same station, two tanks draining. Right: the sector in which our own exhaust would corrupt our own measurements.

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TECHNICAL APPROACH

Technologies to be used

- Python · pandas · NumPy · pvlib (PV physics) · LightGBM (quantile regression)
- PuLP + HiGHS mixed-integer solver · SciPy · Parquet cache · pytest
- ERA5 reanalysis (Copernicus, via an open archive API — no credentials, cached offline)
- Offline single-file HTML console (Plotly inlined) · Modbus + MQTT setpoint interface

Methodology and process for implementation

- Closed-loop simulation of a 210-day season (2023-02-01 to 2023-08-30), re-optimised on a 36 h rolling horizon
- Four controllers on identical weather: fixed schedule, tuned rule-based, ours, perfect-foresight oracle — plus a point-forecast ablation
- Stress tests: four-day blizzard, 48 h genset failure, 12 h badly wrong forecast

Closed loop: every controller is settled by the same physical resolver (min-load bands, SoC limits, tier shedding) on identical weather

Every stage is implemented and runs end to end on one command; the whole pipeline works with the network down, which is the station's real operating condition.

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FEASIBILITY AND VIABILITY

Analysis of the feasibility

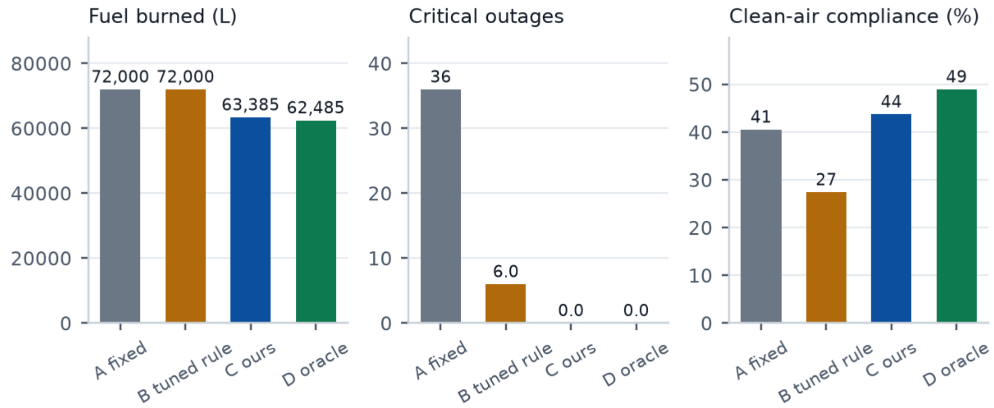
- Already built and running: no hardware, no data collection, no annotation
- MILP solves in ~667 ms per step — an edge laptop is enough
- Forecast skill on an unseen year: load 2.2% MAPE, solar 12.1%, wind 20.0%
- p90 forecast band covers 89% of hours — calibrated, which is what the reserve rule consumes

Potential challenges and risks

- No public electrical-load data exists for ANY polar research station
- Maitri II generator, PV and turbine ratings are not yet published
- The team writes both the simulator and the controller
- Margin over a well-tuned rule-based controller can be modest

Strategies for overcoming these challenges

- Load synthesised bottom-up from physics, never drawn; calibrated to published Antarctic station fuel-demand shapes
- All 64 parameters carry source, confidence and a sensitivity range, shown in an Assumptions tab in the demo
- Honest baselines: a tuned rule-based controller and a perfect-foresight oracle bound
- Framed as a sizing and operating-policy explorer for Maitri II, turning the unknown ratings into the use case



Same weather, same twin. Ours closes 91% of the gap between the tuned baseline and a perfect-foresight bound, at 0 critical outages against 6.

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IMPACT AND BENEFITS

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Potential impact on the target audience

- NCPOR station engineers at Maitri and Bharati: decision support for a fatigued operator in polar night, not a replacement
- Maitri II — approved, ~Rs 2,000 crore, targeted 2029, explicitly a green station: this is an operating policy and a sizing tool for a station still on the drawing board
- Directly transferable to Himalayan, island and forward defence outposts on annual resupply

Benefits of the solution

- Economic — 8,615 L less diesel over the modelled season; fuel delivered by ice-class ship costs a multiple of its pump price
- Operational — critical outages cut from 6 to 0 on identical weather; unserved critical energy 328.4 → 0.0 kWh
- Scientific — clean-air sampling windows protected 27% → 44%, so the station's own record stays usable
- Environmental — less combustion and fewer fuel movements in an Antarctic Treaty protected area; renewables actually used 92% of what they generate
- Strategic — an Indian-built operating policy for Indian polar assets, not a licensed foreign EMS

8,615 L

diesel saved per season
vs a tuned rule-based controller

6 -> 0

critical outages in the same
weather, same station

27% -> 44%

of clean-air sampling windows
kept free of own exhaust

667 ms

per optimisation step
on laptop-class hardware

Measured over the simulated season on identical weather; every figure is regenerated from results/run.json, never typed in.

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RESEARCH AND REFERENCES

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Polar energy literature

- de Witt, Chung & Lee (2024). Mapping Renewable Energy among Antarctic Research Stations. Sustainability 16(1), 426. doi:10.3390/su16010426
- Modeling Hybrid Renewable Microgrids in Remote Northern Regions. Energies (2025) 18, 5827. doi:10.3390/en18215827
- Learning from Arctic Microgrids: cost and resiliency with hydrogen and battery storage. Sustainability (2025) 17, 5996. doi:10.3390/su17135996
- Parent & Ilinca (2011). Anti-icing and de-icing techniques for wind turbines. Cold Regions Science and Technology 65, 88–96
- MPC-based energy management for an isolated electro-thermal microgrid. Energy Conversion and Management (2024)

Data and tooling

- ERA5 reanalysis, Copernicus Climate Change Service / ECMWF — hourly, 9 years at 70.77 S, 11.73 E (Maitri, Schirmacher Oasis)
- pvlib-python for irradiance transposition and cell temperature
- HiGHS mixed-integer solver; LightGBM quantile regression
- SCAR READER, AMRC automatic weather stations, BSRN radiation — cross-checks

Station and programme sources

- NCPOR / COMNAP Antarctic Station Catalogue: Maitri (1989), Bharati (2012)
- Public reporting on Maitri II approval: ~Rs 2,000 crore, target January 2029, wind and solar powered, waste-heat recovery, unmanned data relay
- SIH 2026 portal record for SIH26061 (MoES / NCPOR, Clean & Green Technology)

Our work

- Code, tests, results and the offline console: github.com/NSUT-SIH-26/NSUT-SIH-DEMO
- One command reproduces every number on these slides from real ERA5 weather